

The Hybrid Manager in Construction: Integrating Financial Acumen and Strategic Foresight in Project-Based Enterprises

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Abstract—This paper investigates the evolving role of the commercial manager within UK construction enterprises, positioned at the intersection of engineering management and commercial project leadership. Through structured interviews with practicing quantity surveyors, commercial managers, and legal specialists, the research delineates a competency framework essential for managing complex, profit-driven engineering projects. Key findings highlight the centrality of cash flow stewardship, risk intelligence, and procurement strategy as core commercial PM functions, supported by analytical reasoning, negotiation prowess, and systems thinking as critical engineering management capabilities. The study further examines pedagogical implications for developing these hybrid competencies within higher education, advocating for experiential learning models—including simulation, role-play, and supervised industry placements—to bridge the gap between technical training and commercial leadership. This paper contributes to the discourse on integrative management practices in project-oriented sectors, offering an evidence-based perspective on cultivating managers who can simultaneously deliver technical integrity, financial viability, and strategic value.

Index Terms—Commercial management, quantity surveying, construction industry, project leadership, competency framework, experiential learning, risk management, procurement strategy.

I. INTRODUCTION

A. The Changing Landscape of Construction Project Management

The United Kingdom construction industry has undergone profound transformation over the past three decades. Market liberalization, intensified competition, and cyclical economic pressures have fundamentally altered the operational environment for construction firms. Where once relationships and recommendations secured work, today competitive tendering dominates, with one interviewee noting that “99% of the work goes to the lowest bidder.” This shift has compressed profit margins to unprecedented levels, forcing organizations to scrutinize every cost and maximize every revenue opportunity [1, 2].

In this challenging context, the traditional role of the quantity surveyor has evolved substantially. The professional concerned primarily with measurement, valuation, and cost control has given way to a more sophisticated actor: the commercial manager, whose remit extends beyond technical accuracy to encompass financial strategy, risk intelligence, and cross-functional leadership. This evolution reflects broader

trends in project-based industries, where technical competence alone no longer suffices for organizational success [3, 4].

The emergence of the commercial manager as a distinct role within construction organizations represents a response to these market pressures. These professionals sit at the intersection of engineering delivery and financial management, responsible for ensuring that projects not only meet technical specifications but also contribute positively to corporate profitability. Their decisions shape cash flow trajectories, influence subcontractor relationships, and ultimately determine whether projects succeed or fail in financial terms [5, 6].

B. The Research Gap

Despite the growing importance of commercial management in construction, the literature has been slow to recognize and define this emerging role. Standard project management frameworks, including those from professional bodies, contain minimal reference to commercial functions. Academic research has similarly neglected this domain, with few studies examining the tasks, skills, and competencies that distinguish effective commercial managers from traditional quantity surveyors [7, 8].

This gap has practical consequences. Educational institutions designing programs for construction professionals lack a clear competency framework to guide curriculum development. Employers seeking to recruit or develop commercial managers have no established benchmarks against which to assess candidates or design development programs. Individual practitioners aspiring to commercial roles lack clarity about the capabilities they need to cultivate [9, 10].

The present study addresses this gap through systematic investigation of commercial management practice in UK construction. By eliciting detailed accounts from practitioners, we develop a comprehensive picture of what commercial managers actually do, the skills they require, and how these capabilities might be developed through education and experience.

C. Research Objectives

This study pursues three primary objectives. First, we seek to define commercial management within the construction industry context, establishing a working definition grounded in practitioner perspectives rather than abstract theorizing. Second, we aim to identify the tasks, skills, and competencies

associated with effective commercial management, differentiating these from traditional quantity surveying functions. Third, we explore the implications of these findings for higher education, examining how commercial management capabilities might be developed through program design and pedagogical innovation [11, 12].

D. Theoretical Framework

Our investigation is informed by competency-based approaches to professional development, which emphasize the integration of knowledge, skills, and attitudes in effective practice. This framework recognizes that professional capability extends beyond technical knowledge to encompass situational judgment, interpersonal effectiveness, and strategic thinking. The competency perspective is particularly relevant to commercial management, where success depends on the ability to navigate complex, uncertain situations requiring both analytical rigor and relational skill [13, 14].

We also draw on experiential learning theory, which posits that meaningful professional learning occurs through cycles of concrete experience, reflective observation, abstract conceptualization, and active experimentation. This framework has particular relevance for understanding how commercial expertise develops over time and for designing educational interventions that accelerate this development [15, 16].

E. Paper Organization

The remainder of this paper proceeds as follows. Section II describes the research methodology, including sample selection, interview protocols, and analytical approach. Section III presents findings organized around key themes: the definition of commercial management, core tasks, essential competencies, and distinctions between contractor and consultant perspectives. Section IV discusses the implications of these findings for educational practice, examining pedagogical approaches including simulation, role-play, and supervised work experience. Section V addresses implementation challenges and institutional considerations. Section VI concludes with recommendations for practice and future research [17, 18].

F. Significance of the Study

This research makes several contributions to theory and practice. For academics, it provides an empirically grounded account of commercial management as an emerging professional domain, filling a notable gap in the construction management literature. For educators, it offers a competency framework to guide curriculum development and pedagogical innovation. For practitioners and employers, it provides benchmarks for recruitment, development, and performance assessment of commercial management talent [19, 20].

II. RESEARCH METHODOLOGY

A. Research Design

The study employed a qualitative research design based on structured interviews with construction industry professionals. This approach was selected for its capacity to capture rich,

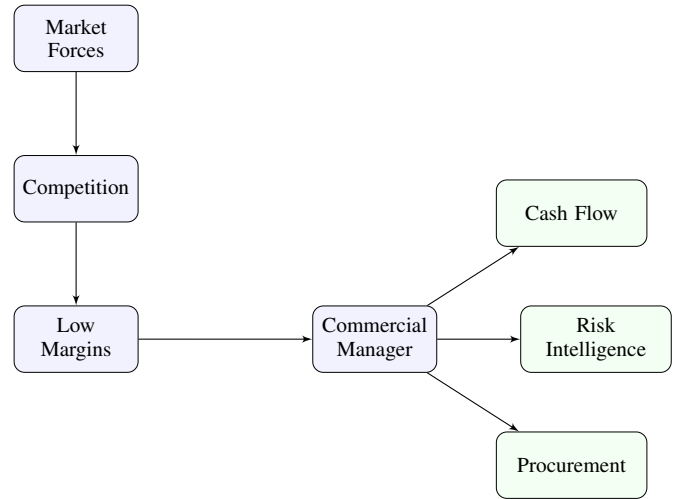


Fig. 1: Market drivers of commercial management.

contextualized accounts of professional practice and to surface tacit knowledge that might not be accessible through survey methods. The interview protocol drew on critical incident technique, asking participants to describe specific situations where they performed effectively or ineffectively and to articulate the behaviors associated with these outcomes [21, 22].

B. Sample Selection

Eighteen participants were recruited for the study, comprising five private quantity surveyors, eleven individuals holding commercial manager positions (though not always with that exact title), and two solicitors specializing in construction law. One solicitor had previously worked as a commercial manager, providing valuable dual perspective. Participants were drawn from leading UK construction organizations including Bovis Construction, Bucknall Austin PLC, Costain, Laing North West, and Sir Alfred McAlpine [23].

The sample was purposively selected to capture diversity of experience and perspective while ensuring all participants had substantial relevant expertise. Commercial manager participants represented both main contractors and specialist subcontractors, while quantity surveyors came from leading private practices. Solicitors were selected for their involvement in construction disputes and contract negotiations.

C. Interview Protocol

Interviews were conducted by an occupational psychologist trained in job analysis techniques, ensuring consistency and depth in data collection. The protocol covered four main areas:

- Definitions and conceptions of commercial management
- Detailed descriptions of tasks performed in the commercial manager role
- Critical incidents illustrating effective and ineffective performance
- Skills and competencies required for successful commercial management

- Views on whether and how commercial management could be taught in universities

Supervisors of interviewees were also interviewed separately to provide confirmatory data and triangulate findings. This multi-source approach enhanced the validity and reliability of the resulting competency framework.

D. Data Analysis

Interview transcripts were analyzed using content analysis techniques. For job analysis components, incidents were coded and categorized to compile comprehensive lists of tasks, skills, and competencies. For definitional questions, key themes were identified through iterative reading and discussion among the research team. Findings were validated through member checking with selected participants and through comparison with existing literature where available [1, 2].

E. Limitations

Several limitations should be acknowledged. The sample, while diverse, was limited to UK-based organizations and may not fully capture international variations in commercial management practice. The cross-sectional design captures current practice but cannot track evolution over time. Reliance on self-report raises potential for retrospective bias, though this was mitigated by supervisor interviews and critical incident technique [3, 4].

TABLE I: Interview Sample Composition

Participant Category	Number
Private Quantity Surveyors	5
Commercial Managers (Contractors)	11
Construction Solicitors	2
Total	18

III. FINDINGS: THE COMMERCIAL MANAGER ROLE DEFINED

A. Context: The Changing Construction Environment

Participants universally described a construction industry transformed by competitive pressure. Over the preceding decade, and particularly during the recessionary period four to five years before the study, the volume of available work had declined while the number of firms tendering had increased. This combination forced margins to razor-thin levels, with one participant noting that tendering returns were “pretty breakneck” compared to earlier periods when work came through recommendation.

This environment fundamentally altered the quantity surveyor’s role. Where previously the focus had been on measurement and valuation with reasonable confidence of eventual profit, the new reality demanded active financial management. As one interviewee explained: “In the good old days QSs measured rates and we had arguments about those rates... but nobody worried about it because the money came in eventually. These days you are generally spending more than you are paid, the focus is absolutely acute.” [5, 6].

B. Defining Commercial Management

When asked to define commercial management, participants converged on a core formulation: “Commercial management is looking after the profits of the company. That is done by keeping costs to a minimum and maximising income.” This definition centers financial stewardship as the primary function, with all other activities subordinate to this objective.

The definition encompasses both operational and strategic dimensions. Operationally, commercial managers ensure that individual projects contribute positively to corporate profitability through careful cost control and revenue maximization. Strategically, they consider the cumulative financial implications of project-level decisions for the enterprise as a whole, recognizing that cash flow from construction activities funds broader corporate operations [7, 8].

C. Core Tasks of Commercial Management

Analysis of task descriptions revealed a coherent set of activities spanning the project lifecycle. At tender stage, commercial managers assess risks embedded in contract documents, evaluate the financial implications of different procurement approaches, and determine appropriate risk contingencies. During procurement, they package works for subcontracting, research subcontractor capabilities, and analyze tender submissions to identify optimal combinations of price and quality.

Throughout project execution, cash flow management occupies central attention. Commercial managers structure payment arrangements so that subcontractors are paid after client receipts, analyze weekly or monthly site reports to track financial performance, and forecast final financial outcomes to enable proactive intervention. They negotiate payments with clients, particularly when variations or changed conditions increase costs, and manage the final account process to maximize recovery [9, 10].

Dispute management represents another significant task area. When subcontractor claims arise, commercial managers investigate underlying circumstances, analyze contractual entitlements, and determine appropriate responses. This may involve negotiation, formal dispute resolution procedures, or in extreme cases, litigation. Throughout, they maintain comprehensive documentation to protect the company’s position.

D. Foundational Competencies

Participants identified four foundational competencies as prerequisites for commercial management. Analytical skills enable interpretation of complex numerical and written information presented in tenders, reports, and contracts. Numerical skills, specifically interpretation rather than mere calculation, allow commercial managers to extract meaning from cash reports, evaluate tender competitiveness, and forecast future financial positions.

Technical quantity surveying skills provide the essential toolkit: measurement, valuation, contract documentation, and cost estimation. Without these fundamentals, commercial managers cannot credibly assess subcontractor claims, evaluate

TABLE II: Core Tasks of Commercial Management

Task Category	Specific Activities
Tender Management	Risk assessment, procurement strategy, contingency determination
Procurement	Work packaging, subcontractor research, tender analysis
Cash Flow Management	Payment structuring, site report analysis, forecasting, client negotiation
Dispute Management	Claim investigation, negotiation, documentation, resolution
Reporting	Financial reporting to senior management, client progress reports

variation impacts, or determine appropriate payments. Logical thinking underpins all other competencies, enabling systematic risk assessment, option evaluation, and decision-making under uncertainty [11, 12].

E. Advanced Competencies

Beyond foundational skills, participants identified several higher-order competencies distinguishing effective commercial managers. Cash management, described as “good housekeeping,” involves applying common-sense financial principles: ensuring subcontractors are paid after client receipts, analyzing payment flows, and forecasting future positions to enable proactive management.

Broad perspective-taking enables commercial managers to see beyond immediate project concerns to understand wider implications. This includes recognizing how project cash flows affect corporate financial health, appreciating that dispute resolution through negotiation may serve long-term interests better than litigation despite short-term financial trade-offs, and anticipating how current decisions may shape future opportunities [13, 14].

Risk assessment operates continuously throughout project lifecycles. Commercial managers identify potential risks in tenders, subcontractor arrangements, and client situations, then develop strategies to minimize exposure. This involves quantifying potential costs, evaluating acceptability, and implementing mitigation measures. Proactivity distinguishes effective risk management: anticipating problems before they materialize rather than reacting to crises [15, 16].

Negotiation skill combines interpersonal effectiveness with strategic thinking. Successful negotiators understand when to concede minor points to secure major advantages, when flexibility serves interests better than rigid adherence to contract terms, and how to build relationships that facilitate future dealings. Communication, both oral and written, ensures that all parties understand agreements, variations, and expectations, minimizing disputes arising from misunderstanding.

F. Private Practice Perspectives

Quantity surveyors in private practice offered complementary perspectives. For them, commercial awareness involved understanding development appraisal—how developers structure financial assessments of projects, including rents, yields, and critical cost drivers. Market knowledge was essential to advise clients on building designs that would attract tenants or buyers. Procurement strategy expertise enabled appropriate advice on whether time, cost, or quality should be the primary criterion for specific projects.

Cash flow management received less emphasis in private practice, though at senior levels, partners needed to understand profit margins and the financial implications of practice decisions. This confirms that commercial awareness operates differently depending on organizational position: contractors focus on project cash flows, while consultants emphasize client financial contexts [17, 18].

G. The Educational Imperative

Despite emphasizing experience as the primary teacher, participants strongly supported including commercial management in university curricula. They acknowledged that graduates with theoretical grounding developed faster and performed better than those entering without conceptual frameworks. However, they stressed that theoretical approaches alone would not suffice; students needed exposure to realistic situations where they could apply concepts, analyze risks, solve problems, and experience the consequences of decisions [19, 20].

Participants suggested various pedagogical approaches: computer simulations enabling students to manage virtual projects, group exercises with role-playing, and visiting speakers from industry. One emphasized that students should set the agenda for guest speakers, encouraging active engagement rather than passive reception. Another noted that while commercial skills should be developed at university, technical quantity surveying fundamentals remained paramount for new graduates [21, 22].

IV. PEDAGOGICAL IMPLICATIONS AND CURRICULUM DESIGN

A. Experiential Learning Foundations

The emphasis on experience as the primary teacher of commercial skills points toward experiential learning approaches in higher education. Kolb’s experiential learning cycle—concrete experience, reflective observation, abstract conceptualization, active experimentation—provides a framework for designing interventions that accelerate development of commercial judgment. Rather than simply transmitting knowledge about commercial management, educators must create conditions where students can practice, reflect, conceptualize, and experiment [23].

Experiential approaches are particularly suited to developing the higher-order competencies identified in this study. Cash flow management, risk assessment, and negotiation cannot be learned solely through lectures; they require situated practice with feedback and opportunities for reflection. The challenge

for educators is to create sufficiently realistic learning environments within institutional constraints [1, 2].

B. Simulation and Gaming

Simulation offers one promising approach. AROUSAL (A Real Organisational Unit Simulated as Life), developed by Lansley, recreates the decision environment of construction companies, modeling corporate strategy, bidding, cash flow management, recruitment, and project staffing. Students make decisions and observe consequences in a risk-free environment, developing understanding of how commercial decisions interact across the enterprise.

Games introduce competitive elements that motivate engagement and highlight trade-offs. When students compete for simulated contracts while managing cash flows and subcontractor relationships, they experience the pressures of real commercial environments. Debriefing sessions following simulations are essential, enabling students to articulate what they learned and connect experiences to theoretical frameworks [3, 4].

C. Role-Play and Group Exercises

Role-play allows students to experience different perspectives in commercial relationships. Taking the role of commercial manager, subcontractor, or client, students negotiate payments, resolve disputes, and make procurement decisions. The interpersonal dynamics of these exercises develop communication and negotiation skills while highlighting how different parties construct meaning from identical situations.

Group exercises build teamwork capabilities essential for commercial practice. When students must coordinate across roles, delegate tasks, and integrate diverse contributions, they develop skills for working in project-based organizations. Feedback from peers and tutors provides developmental insight unavailable in individual assignments [5, 6].

D. Supervised Work Experience

Supervised work experience, embedded in sandwich degree programs, provides extended opportunities for learning in authentic settings. The Commercial Management and Quantity Surveying program at UMIST requires 46 weeks of supervised experience, enabling students to observe and participate in real commercial activities. However, placement learning does not automatically occur; it must be structured to maximize developmental value [7, 8].

Learning contracts specifying what students will achieve during placements enhance focus and intentionality. Involving workplace mentors in contract negotiation builds commitment and clarifies expectations. Formal assessment of placement learning signals its importance and ensures comparable experiences across students. Personal and Academic Development Plans help students reflect systematically on their experiences and plan future learning [9, 10].

TABLE III: Pedagogical Approaches for Commercial Management Development

Approach	Key Features
Simulation	Computer-based decision environments modeling commercial dynamics
Gaming	Competitive elements highlighting trade-offs and consequences
Role-Play	Perspective-taking in commercial relationships and negotiations
Group Exercises	Collaborative problem-solving developing teamwork capabilities
Supervised Work Experience	Extended authentic practice with structured reflection
Visiting Speakers	Industry practitioners sharing contextualized experience

E. Curriculum Integration Challenges

Integrating experiential learning into traditional university programs faces multiple challenges. Conservative faculty and students may resist unfamiliar approaches, perceiving project work as higher-risk than traditional methods. Assessment of experiential learning requires innovative approaches that capture process as well as product, individual as well as group contribution.

Staff roles shift from knowledge transmitters to learning facilitators, requiring new skills and orientations. Faculty must provide resources, critical feedback, and assessment while relinquishing control over learning processes. This transformation requires institutional support and faculty development initiatives [11, 12].

V. IMPLEMENTATION CONSIDERATIONS AND INSTITUTIONAL FACTORS

A. Organizational Context for Learning

The development of commercial capability extends beyond formal education into workplace learning. Construction organizations must create environments where junior staff can develop commercial judgment through guided experience. This requires senior practitioners who model effective practice, provide feedback, and progressively delegate responsibility. Mentoring relationships accelerate learning by providing access to experienced practitioners' tacit knowledge [13, 14].

Organizations also need systems that capture and disseminate commercial learning. Post-project reviews, knowledge databases, and communities of practice enable learning from across the organization. Without such systems, lessons remain individual rather than organizational, and each generation of commercial managers must reinvent practices [15, 16].

B. Professional Body Engagement

Professional bodies have significant influence on commercial management development through accreditation standards, continuing professional development requirements, and

qualification pathways. The Royal Institution of Chartered Surveyors' identification of core skills—analysis, appraisal, communication, documentation, management, quantification, synthesis—provides a framework that partially overlaps with commercial competencies [17, 18].

However, professional frameworks may need updating to fully capture commercial management's distinctive requirements. The emphasis on cash flow stewardship, risk intelligence, and procurement strategy identified in this study suggests areas where current frameworks could be enhanced. Professional bodies also have roles in developing continuing education programs for experienced practitioners seeking to enhance commercial capabilities [19, 20].

C. University-Industry Partnerships

The UMIST program's consortium of sponsoring contractors exemplifies productive university-industry collaboration. Sponsors provide input to curriculum design, offer placement opportunities, and engage with students through guest lectures and project supervision. This engagement ensures that programs remain relevant to industry needs while exposing students to practitioner perspectives.

Such partnerships require sustained investment from both parties. Universities must design programs flexible enough to accommodate industry input while maintaining academic rigor. Industry partners must commit time and resources to engagement activities without immediate return. Successful partnerships recognize mutual benefits: universities gain relevant curricula and placement opportunities; industry gains graduates prepared for productive contribution [21, 22].

D. Assessment Challenges

Assessing commercial capability presents particular challenges. Traditional examinations measure knowledge but cannot capture judgment in action. Coursework assignments can simulate aspects of commercial practice but lack the complexity and consequences of real situations. Workplace assessment, while authentic, raises consistency and comparability concerns.

Multiple assessment methods may be needed to capture different dimensions of commercial capability. Simulations can assess decision-making under uncertainty. Role-plays can evaluate negotiation and communication skills. Reflective portfolios can demonstrate integration of learning across experiences. Workplace assessments by qualified mentors can provide evidence of authentic practice. Combining methods provides triangulation while acknowledging that no single approach fully captures commercial competence [23].

VI. CONCLUSION AND RECOMMENDATIONS

A. Summary of Contributions

This study has provided the first systematic investigation of commercial management as a distinct role within UK construction. Through structured interviews with practitioners, we have developed a competency framework encompassing foundational skills (analytical, numerical, technical, logical),

advanced capabilities (cash management, broad perspective, risk assessment), and integrative competencies (negotiation, communication, team working). We have distinguished contractor and consultant perspectives, showing how commercial emphasis varies by organizational position.

We have also explored implications for education and development, identifying experiential learning approaches aligned with how commercial capability develops in practice. Simulation, role-play, group exercises, and supervised work experience offer ways to accelerate development within higher education. Implementation requires attention to curriculum design, faculty development, assessment innovation, and university-industry partnership [1, 2].

B. Recommendations for Practice

For educational institutions, we recommend integrating experiential learning throughout commercial management programs. Curricula should include progressively complex simulations, structured work placements, and opportunities for reflective practice. Faculty development should support transition from traditional teaching to learning facilitation. Assessment should capture multiple dimensions of commercial capability through diverse methods.

For construction organizations, we recommend systematic approaches to commercial talent development. Recruitment should assess both foundational skills and potential for advanced competencies. Development programs should combine formal learning with guided workplace experience. Mentoring should connect junior staff with experienced practitioners. Knowledge systems should capture and disseminate commercial learning across the organization [3, 4].

For professional bodies, we recommend updating competency frameworks to fully reflect commercial management's distinctive requirements. Continuing professional development should include offerings addressing cash flow management, risk intelligence, and procurement strategy. Qualification pathways should recognize commercial capability as a distinct professional domain [5, 6].

C. Recommendations for Policy

Government and industry bodies should support initiatives that strengthen commercial capability in construction. Funding for university-industry partnerships, support for work placement programs, and recognition of commercial management in skills strategies would signal the importance of this domain. Research funding should support further investigation of commercial practice and its development [7, 8].

D. Future Research Directions

Several directions for future research emerge from this study. Comparative research across countries and construction systems would reveal how commercial management varies with institutional context. Longitudinal studies tracking commercial manager development would illuminate learning trajectories and critical experiences. Intervention studies evaluating alternative pedagogical approaches would provide evidence for educational design [9, 10].

Research should also examine commercial management in related project-based industries: engineering, infrastructure, energy, and technology. Similar pressures toward financial stewardship and strategic foresight may be creating parallel roles requiring comparable competencies. Cross-sector comparisons would reveal domain-specific variations and generic principles [11, 12].

Finally, research should investigate how commercial capability relates to organizational performance. Does stronger commercial management correlate with higher profitability, lower dispute rates, or more successful projects? Such evidence would strengthen the business case for investing in commercial talent development [13, 14].

E. Concluding Reflection

The emergence of commercial management as a distinct role reflects fundamental changes in the construction industry's operating environment. Intensified competition and compressed margins have made financial stewardship as important as technical delivery. The commercial manager embodies this evolution, positioned at the intersection of engineering management and financial strategy.

Developing commercial capability requires rethinking both education and practice. Universities must move beyond technical training to cultivate judgment, perspective, and relational skill. Construction organizations must create environments where commercial learning can occur through guided experience. Professional bodies must recognize and support this emerging domain.

The hybrid manager—combining technical grounding, financial acumen, and strategic foresight—represents the future of project leadership in construction and beyond. This study provides foundation for understanding and developing this critical capability [15–23].

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